

Learner migration 2027-2035

Region IV-A | Calculation explanation and readable agent report

Run: 2026-10-07T02-26-24.718Z. Baseline: **3,822,536 learners**, SY 2025-2026.

Ang output ay sensitivity scenario gamit ang observed enrollment baseline. Ang annual migration rates ay assumptions; hindi pa sila fitted o validated learner-migration forecasts.

Paano binabasa ang baseline

Ang year ay ending year ng school year. Baseline 2026 = SY 2025-2026; output 2027 = SY 2026-2027; output 2035 = SY 2034-2035. Ang index na 100 ay normalization ng actual baseline, hindi 100 learners. [1, 2]

Formula at assumptions

$$\text{Learners}(y) = \text{round}(B \times (1 + r / 100)^{(y - 2026)})$$

B = 3,822,536; r = assumed annual net migration contribution; y = school-year ending year. Low: -0.5%; base: 0%; high: +0.5%. Compounded ang pagbabago taun-taon. [1, 3]

Ang rates ay hindi kinuha sa PSA general-population migration percentages, IMPACT findings, o DepEd enrollment changes. Births, cohort aging, progression, dropout at international migration ay held fixed/excluded para makita ang migration assumption lamang.

Calculation citations: [1] calculation.json; [2] deped-source.json; [3] run-migration-agents.mjs. Full reference details are at the end.

Calculation: worked examples

2027 low: $\text{round}(3,822,536 \times (1 + (-0.5 / 100))^1) = \mathbf{3,803,423 \text{ learners. [1]}}$

2035 high: $\text{round}(3,822,536 \times (1 + (0.5 / 100))^9) = \mathbf{3,998,031 \text{ learners.}}$ Ito ay +175,495 kumpara sa baseline, conditional sa fixed assumption. [1]

School year	Low (-0.5%)	Base (0%)	High (+0.5%)
2026-2027	3,803,423	3,822,536	3,841,649
2027-2028	3,784,406	3,822,536	3,860,857
2028-2029	3,765,484	3,822,536	3,880,161
2029-2030	3,746,657	3,822,536	3,899,562
2030-2031	3,727,923	3,822,536	3,919,060
2031-2032	3,709,284	3,822,536	3,938,655
2032-2033	3,690,737	3,822,536	3,958,348
2033-2034	3,672,284	3,822,536	3,978,140
2034-2035	3,653,922	3,822,536	3,998,031

Paano pinapanatili ang domestic total

Other domestic learners = 25,910,491 - modeled regional learners. Halimbawa, kapag nadagdagan ang destination, ibinabawas ang parehong bilang sa pooled counterpart. PSO is excluded. Walang specific origin-region allocation na na-estimate sa model. [1, 2]

Hindi ito full enrollment forecast. Ang stock changes sa observed history ay maaaring mula sa cohort, reporting, o geography changes. Ang residence migration, school transfers, at commuting ay magkakaibang measures. PSA projections may already incorporate migration assumptions; kailangan ang methodological check bago magdagdag ng migration adjustments. [4]

Table and calculations: calculation.json from the identified run. Source citations below; public sources supply context, not the numerical rates.

Readable research and source report

Formatted from report.md for the same run. The generated narrative is an analytical interpretation; the calculation and observed baseline above are authoritative for numeric values.

ATLAS: Region IV-A learner migration scenario, 2027-2035

Observed baseline and inspected data

Actual enrollment baseline: **3,822,536 learners**, SY 2025-2026, from [depedprototype/depedprot/regional_profiles/Region IV-A:enrollment_total_2526](#). The index normalizes this observed count to 100 at school-year ending year 2026. Year 2027 represents SY 2026-2027; year 2035 represents SY 2034-2035.

The connector inspected regional_profiles, 35 selected-region historical KPI records (2018-2019 to 2024-2025), national_kpi_timeline, and 79 raw/materialized-view catalog entries. It is no longer based only on department_orders. Snapshots and document paths are in deped-source.json.

Regional enrollment stock: SY 2023-2024 = 3,951,663; SY 2024-2025 = 3,871,761; SY 2025-2026 = 3,822,536. Enrollment changes alone do not identify migration: reporting, geographic boundaries, cohort changes, progression and exits may also contribute.

Historical education indicators: SY 2024-2025

Level	GER (%)	NER (%)	Retention (%)	Transition (%)	School-leaver (%)
Elementary	96.17	88.02	99.22	98.21	0.54
JHS	91.05	84.23	97.08	97.74	2.93
JHS to SHS	94.39	87.02	96.89	98.28	3.02
Kindergarten	85.33	75.97	Not supplied	98.39	Not supplied
SHS	95.52	74.93	Not supplied	Not supplied	Not supplied

Values above are percentage values copied from kes_historical and rounded to two decimals for presentation. Profile retention/transition fields use proportions; their units must be normalized before combining data. Missing values remain missing. These indicators provide education context; they are not annual learner migration rates and are not inputs to the current calculator.

Research evidence

The IMPACT report supplies qualitative context, not a calibrated learner-transfer rate. The following verbatim excerpts passed chunk/page validation. Historical policy statements describe the study publication period and do not establish current policy status.

- PDF page 17, chunk p017-c01: "Internal migration in the Philippines can be described as largely rural to urban and rural to rural, it is also most commonly carried out by women and youth (Gultiano and Xenos 2004; Quisumbing and McNiven 2005). Migration flows are mainly interprovincial, with the National Capital Region (NCR) and surrounding areas as the prime destination (Quisumbing and McNiven 2005). Results of the 2018 NMS shows that the percentage of Filipinos with internal migration experience ranges from 4% in Central Luzon to 20% in Bicol, Eastern Visayas (22%), and ARMM (22%)."

- PDF page 73, chunk p073-c01: "At present, there is no policy that explicitly mentions and addresses migration and displacement in the context of climate change (Vinke 2020). It is important to prioritise an improved understanding of the climate-migration nexus that includes monitoring the trends and impacts of rapid and slow onset events, and their relationship to human mobility through displacement, immobility and migration."

- PDF page 18, chunk p018-c01: "Education and social networks are some of the 'pull factors' while unemployment, low wages, low profitability in farming, lack of infrastructure facilities in the rural areas are some of the reasons why people leave their villages (Paris et al. 2009). Family pressure and family ties, and to a lesser extent prior migration experience, economic resources, marital status and age are also considered important factors to a person's migration intentions (De Jong et al. 1985). Migration is a common livelihood strategy of poor farming households which helps to smooth seasonal income fluctuations and earn extra cash to meet contingencies or increase disposable income (Paris et al. 2009)."

- PDF page 69, chunk p069-c01: "With regard to social networks, people arriving in destination areas may need to build relationships and social capital. Since people in destination areas often come from different backgrounds and culture, it would be important to strengthen communication and interaction to avoid social tensions. The capacity of hosts to receive migrants and displaced persons would need to be enhanced to facilitate local integration (UN 2020). Culture is an important factor that influences how well arriving populations are able to adapt to new locations."

PSA, BetterGov and ASEAN

- PSA: web_retrieved; 6 source citations retained. Connection type: Public web search; not a statistical API integration.

- BetterGov: web_retrieved; 4 source citations retained. Connection type: Public web search; not a statistical API integration.

- ASEAN: web_retrieved; 7 source citations retained. Connection type: Public web search; not a statistical API integration.

PSA publishes 2020 Census historical origin-destination migration tables (2015-2020) and population projections. Those household/population measures require population, age, geography and time alignment before estimating enrolled-learner flows. Place-of-school measures may represent commuting rather than residential migration or a school transfer. Retrieved URLs appear below; the present run has not loaded their statistical tables into the calculator. BetterGov is a community portal; upstream PSA provenance must be retained. ASEAN sources offer regional context, not a measured Philippine learner-transfer series in this calculation.

Calculation and 2027-2035 results

Assumed annual net migration contributions: **-0.5%, 0%, 0.5%**. These are illustrative defaults, not empirically estimated. Formula: $\text{learners}(y) = \text{round}(\text{baseEnrollment} \times (1 + r/100)^{(y-2026)})$. Neither PSA flows nor research statistics were used to estimate r . All nonmigration changes are held fixed.

School year	Low learners	Base learners	High learners
2026-2027	3,803,423	3,822,536	3,841,649
2027-2028	3,784,406	3,822,536	3,860,857
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Internal transfers are conserved: other domestic learners = 25,910,491 - modeled regional learners. PSO is excluded. The other domestic regions form a pooled counterpart; the model does not identify individual origin or destination counterparts. This is a sensitivity scenario conditional on the assumed rates, not a calibrated migration forecast or a full demographic enrollment projection.

Revised conclusion and next steps

There is usable existing DepEd enrollment/KPI history and PSA migration/population evidence. The remaining task is to align their measures and estimate the learner-specific component, rather than assume data is absent. This run has not confirmed a direct learner origin-destination flow field in the inspected catalog entries; underlying source artifacts and uncatalogued tables remain to be investigated. Missing retrieval does not establish dataset absence.

For calibration: retrieve the PSA statistical tables, reconcile geographic boundaries and age groups with DepEd enrollment, distinguish commuting from transfers and residential moves, fit the learner-specific component, and evaluate against held-out observations. Climate indicators are needed when estimating climate-related effects. Check whether any adopted PSA projection already includes migration assumptions before adding a separate migration adjustment.

Calculation citations: calculation.json; observed snapshots: deped-source.json; research provenance: research-evidence.json and IMPACT-Study-Final-web-page-2.pdf; arithmetic implementation: scripts/run-migration-agents.mjs. The original LLM explanation is retained separately as a draft in explanation-agent.json; this report replaces its unverified KPI summaries and broad current-policy assertions with snapshot values and bounded evidence.

Citations and provenance

[1] ATLAS calculation.json. Scenario parameters, formulas and 2027-2035 values; same run shown on page 1. Numerical rates are illustrative defaults, not source estimates.

[2] ATLAS deped-source.json. Read-only snapshot of depedprototype/depedprot: regional_profiles, selected-region kes_historical, national_kpi_timeline, raw_catalog and mv_catalog. Baseline field: regional_profiles/Region IV-A/enrollment_total_2526 (for this run).

[3] ATLAS scripts/run-migration-agents.mjs. Deterministic regionalScenario calculation, rounding and pooled domestic conservation checks.

[4] ATLAS public-sources.json and report.md. Public web retrieval and agent interpretation. Underlying public-source links appear below.

Research study

Ayeb-Karlsson, S. and Uy, N. Internal Migration in the Philippines: Adaptation to Climate Change (IMPACT), final report. Local source: IMPACT-Study-Final-web-page-2.pdf. Evidence pages/chunks used in this run:

PDF page 17, p017-c01; PDF page 73, p073-c01; PDF page 18, p018-c01; PDF page 69, p069-c01

Retrieved public references

PSA: **Migration and Overseas Workers 2020 Census of Population and Housing (2020 CPH) | Philippine Statistics Authority**

https://psa.gov.ph/content/migration-and-overseas-workers-2020-census-population-and-housing-2020-cph?utm_source=openai

PSA: **Highlights of the Region IV-A (CALABARZON) Population 2020 Census of Population and Housing (2020 CPH) | Philippine Statistics Authority**

https://psa.gov.ph/content/highlights-region-iv-calabarzon-population-2020-census-population-and-housing-2020-cph?utm_source=openai

PSA: **Republic of the Philippines**

https://psa.gov.ph/system/files/dhsd/Press%20Release%20%282020%20CBPP%20Subnational%29.pdf?utm_source=openai

PSA: **2020 CENSUS-BASED NATIONAL POPULATION PROJECTIONS**

https://psa.gov.ph/sites/default/files/dhsd/Statistical%20Tables%20CBPP_0.pdf?utm_source=openai

PSA: **Migration and Overseas Workers**

https://psa.gov.ph/system/files/phcd/2023-01/Special%2520Release%2520on_Migration%2520and%2520Overseas_Worker_09Jan2023_PMMJ_CRD-signed.pdf?utm_source=openai

BetterGov: **Total Population, Household Population, Number of Households and Average Household Size by Region, Province, and Highly Urbanized City: Philippines, 2024 · Philippine Data Explorer**

https://statistics.bettergov.ph/datasets/a37b2e8af4d42c3d0fbc?utm_source=openai

BetterGov: **Philippine Statistical Yearbook statistics · Philippine Data Explorer**

https://statistics.bettergov.ph/search?topic=Philippine+Statistical+Yearbook&utm_source=openai

BetterGov: **Classification · Philippine Data Explorer**

https://statistics.bettergov.ph/classification?utm_source=openai

BetterGov: **TABLE 9.4 Number of Registered Live Births and Rates by Region (Usual Residence of Mother): 1992 to 2023 (Rates per 1,000 population) · Philippine Data Explorer**

https://statistics.bettergov.ph/datasets/cb434367ac4d69d087a7?utm_source=openai

ASEAN: **TABLE 1.1 Population, Land Area, Density by Region and Province: CENSUS YEARS 2000 to 2024 · Philippine Data Explorer**

https://statistics.bettergov.ph/datasets/ad916a257572af037d2b?utm_source=openai

ASEAN: **Regional Philippine statistics · Philippine Data Explorer**

https://statistics.bettergov.ph/regional?utm_source=openai

ASEAN: **<table id="e1">**

https://cdn.aseanstats.org/public/docs/metadata/sits/2021/SITS_Metadata_PH.pdf?utm_source=openai

ASEAN: **ASEAN SUSTAINABLE**

https://www.aseanstats.org/wp-content/uploads/2020/11/ASEAN-Sustainable-Development-Goals-Indicators-Baseline-Report-2020-web.pdf?utm_source=openai